

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 26

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Negative feedback responds to any change by re-establishing the previous equilibrium (1) • Only large unusual events overwhelm the feedback so a new equilibrium forms (1) • Example: methane hydrate release at the end of the Permian causing climate change and mass extinction (1)	3
01.2	<i>1 mark for each point.</i> • In a diverse system each component is relatively less important so the loss / change of one has a small effect; alternative food sources and relationships buffer the change (1) • Natural: coral reefs and tropical rainforests with complex food webs are stable; simple polar / desert systems fluctuate (1) • Human: agroecosystems focus on a few crops and remove detritivores, pollinators and predators so are vulnerable to pests, disease, weather (1) • Diverse technical systems: a society using a range of renewable energy resources is more secure than one relying on a single source (1)	4
01.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Melting permafrost releasing methane (1) • Reduced albedo as ice / snow melt (1) • Methane hydrate release (1) • Forest and peat fires (1) • Faster soil decomposition (1) • Ocean acidification reducing sequestration (1) • Cirrus cloud formation (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	

Question	Marking guidance	Marks
02.1	<i>1 mark.</i> • Direct measurements of past conditions cannot be taken, so indirect evidence that is related to the condition must be used (1)	1
02.2	<i>1 mark for each point.</i> • Age / date of the sample (radioisotope dating) (1) • Past temperatures (1) • Composition of the past atmosphere (eg CO ₂ / methane concentrations) (1) • Temperature: more magnesium is incorporated into calcite at higher temperatures (1)	4
02.3	<i>1 mark for each point.</i> • Movement of continents changed the position of land masses / latitude and therefore the climate and habitats present (1) • Ocean currents redistribute heat and nutrients; changes in currents (eg as continents moved) altered climates, so past conditions cannot be explained without knowing how the systems worked (1)	2
02.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • There was no oxygen / ozone layer so high levels of UV radiation reached the land surface; water absorbed the UV and protected organisms in the oceans (1) • Warm temperatures around the vents and a rich mix of chemicals / increasingly complex organic molecules in the sea made biological reactions more likely (1) • The atmosphere contained toxic gases such as ammonia; the early Archaea were anaerobic so did not need oxygen (1) • Water acted as the solvent for reactions / transport (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-9; AQA spec 3.1.1.1-3.1.1.3.	
03.1	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Hedgerows (1) • Hay meadows (1) • Grazed moorland / heathland (1) • Chalk grassland (1)	2
03.2	<i>1 mark for each point.</i> • Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive (1) • Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers (1)	2
03.3	<i>1 mark.</i> • The introduced (non-indigenous) predator may eat other species as well as the pest (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	
04.1	<i>1 mark for each point.</i> • In-situ: conserving species in their natural habitat with the whole community; ex-situ: away from the natural habitat, eg zoos, botanic gardens, seed banks, captive breeding (1) • Needed when conservation in the wild is impossible or insufficient to prevent extinction (habitat gone, population too small, poaching) (1)	2

Question	Marking guidance	Marks
04.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Is the wild population threatened? (1) • Is there a genetically diverse captive population? (1) • Is in-situ conservation failing? (1) • Is keeping a captive population realistic (habitat size, food, relationships, cost)? (1) • Is release likely to be successful now or in future (suitable habitat, threats removed)? (1)	4
04.3	<i>2 marks for each point (partial credit for a partly correct point).</i> • Blue whale: habitat size far too large to provide; food (huge quantities of krill / plankton) cannot be supplied; cost (2) • Large blue butterfly: complex inter-species relationship - larvae are taken into ant nests where they overwinter feeding on ant grubs; requires specific ant species and thyme food plant (2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-34; AQA spec 3.1.2.3.3.	
05.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Photosynthesis and aerobic respiration balance O₂ and CO₂ in a dynamic equilibrium; evapotranspiration adds water vapour forming clouds and rain; microbes drive the nitrogen and other biogeochemical cycles so wastes do not accumulate and nutrients are recycled (2) • Detritivores, fungi and bacteria decompose dead organic matter releasing nutrients and forming humus that binds soil; organic acids weather rock forming new soil; roots bind soil, reducing erosion and regulating water flow / flooding (2) • Species provide food, pollination (bees, moths, birds, bats - more reliable than wind, especially in forests), seed dispersal (animals carry larger seeds to suitable sites, faeces as fertiliser, gut passage stimulates germination) and habitat provision (trees as nest sites and canopy microclimate; mollusc shells for hermit crabs) (2)	6
05.2	<i>1 mark for each point.</i> • Its nectar is at the end of a 30 cm tube reached only by the long tongue of the sphinx moth, which carries pollen only between orchids of the same species (1) • Advantage: no pollen wasted on other species and the moth has an exclusive food source; risk: if either species declines the other cannot survive (1)	2
05.3	<i>1 mark for each point.</i> • They are not ruminants so seeds are not destroyed by repeated chewing / pass through intact; they eat the fruit of most tree species (1) • Not territorial so they carry seeds over large distances / areas (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 18-19; AQA spec 3.1.2.1.3.	

Question	Marking guidance	Marks
06.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Demersal trawl: seabed species - cod, haddock, plaice, shrimps, scampi; chains / weights damage the seabed (mix aerobic and anaerobic layers, destroy sea fans, corals, seagrass); high by-catch in mixed fisheries; high fuel use (2) • Purse seine: shoaling pelagic species - tuna, sardines, herring, anchovy; can remove whole shoals; dolphins associated with tuna shoals may be caught (2) • Pelagic long-line: tuna, squid; albatrosses (long-lived, low breeding rate) drown on baited hooks; sharks and turtles caught (2)	6
06.2	<i>1 mark for each point.</i> • Shrimp trawls use very small mesh so crabs, molluscs, sea urchins, starfish, juvenile fish cannot escape - by-catch can be 99% of the catch (1) • Drift nets are non-selective curtains near the surface catching whales, dolphins, turtles, sharks (1)	2
06.3	<i>1 mark for each point.</i> • Lost or discarded nets and traps continue to catch and kill organisms; dead animals act as bait attracting more (1) • Biodegradable ropes / panels so lost traps fall apart; radio transmitters on traps so they can be recovered; net recovery schemes (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 340-343; AQA spec 3.5.2.2.	
07.1	<i>1 mark for each point.</i> • Primary: succession on a newly created habitat with no life / no soil (bare rock, new pond, sand) (1) • Secondary: succession restarting after an existing community is disturbed (tree felling, ploughing, fire, end of grazing) (1) • Faster because soil already exists and many seeds / roots / organisms are still present so there are fewer delays for colonisation (1)	3
07.2	<i>1 mark for each point.</i> • Human activity (grazing, mowing, burning, coppicing) regularly removes the later successional species so the climatic climax cannot form (1) • The stable community that develops under this repeated disturbance is a plagioclimax (1) • Examples: chalk grassland / hay meadows maintained by grazing or mowing; lowland heath by burning; coppiced woodland; Machair grassland; Broads grazing marshes (1)	3
07.3	<i>1 mark for each point.</i> • If the traditional activity stops, secondary succession returns the habitat to scrub / woodland and the plagioclimax species are lost (1) • Rainforest and coral reefs are climax communities so conservation aims to minimise human impact (1)	2
07.4	<i>1 mark for each point.</i> • Extreme environments (deserts, polar) are dominated by abiotic factors; few species tolerate them; populations fluctuate and food webs are simple so changes in one food species have large effects (1) • Stable environments allow many species to evolve inter-species relationships; populations are regulated by biotic factors; complex food webs give stability (rainforests, reefs) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 76-77; AQA spec 3.1.3.3.	

Question	Marking guidance	Marks
08.1	<i>1 mark for each point.</i> - t-test (1) - The data are measured (continuous) values and two means are being compared (1) - The data are (assumed to be) normally distributed (1)	3
08.2	<i>1 mark.</i> Use the Mann-Whitney U test instead (compares medians) (1)	1
08.3	<i>1 mark for each point.</i> - There is a 2% probability that the difference occurred by random chance (1) - As this is less than 0.05 the difference is statistically significant / the null hypothesis is rejected (1)	2
08.4	<i>1 mark for each point.</i> - The ranges of values (mean plus or minus SD) do not overlap: woodland up to 1.6 and clearing down to 2.4 (1) - About 68% of values lie within one SD of each mean so the two data sets are clearly separated (1)	2
08.5	<i>1 mark for each point.</i> - Significance shows the difference is unlikely to be due to chance, not that a causal relationship exists (1) - Another variable (eg topography / aspect of the sites) could cause both; a controlled experiment or further evidence is needed (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
09.1	<i>2 marks for each point (partial credit for a partly correct point).</i> - Selective breeding: parents chosen for desirable genetic characteristics (Limousin muscle growth, Merino wool, Cheviot hardiness); combines genes but characteristics of offspring not fully predictable; many generations needed; risk of inbreeding (2) - Asexual reproduction (cuttings, runners, tissue culture, cloning): offspring genetically identical to the parent so characteristics are predictable and survival high; but no variation so no improvement and vulnerability to the same disease; cloning of animals (nucleus transfer into an enucleated egg, surrogate) still developing (2) - Cross-breeding: two breeds combined to give a mix of characteristics with hybrid vigour (heterosis) and lower inbreeding risk, eg Zebu (heat tolerant) x Ayrshire (high milk yield) (2)	6
09.2	<i>1 mark for each point.</i> - AI: semen from a male with desirable characteristics can fertilise many more females than natural mating; frozen semen stored and transported easily and used after the male dies (1) - Embryo transfer: FSH stimulates a superior female to release many eggs, which are fertilised (IVF) with sperm from a superior male (1) - Each embryo is implanted in a different surrogate female (1) - So the donor produces many offspring per year rather than one per pregnancy (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-323; AQA spec 3.5.1.2.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Protected areas: Korup NP Cameroon (1260 km², 600 tree species, endemic <i>Ancistrocladus korupensis</i> with anti-HIV / antimalarial alkaloids); Tumucumaque NP Brazil (39 000 km², adjoining French Guiana reserve); Alto Maues; Rio Bravo Belize funded by charities, companies and ecotourism • Debt for nature swaps; sustainable exploitation: selective logging of high-value timber (mahogany) funding conservation; ITTO aims (largely unmet); CITES listing of timber species (Honduras mahogany Appendix II) • Ecotourism income (but visitor damage in Costa Rica, Malaysia); community projects and employment; carbon sequestration payments • Effectiveness: protected areas work where enforced but face illegal logging, mining, poaching, roads, conflict; drivers (population growth, demand for beef, soya, palm oil, timber, minerals) originate outside the forest; climate change cannot be controlled locally • Judgement: combination of large protected areas, economic alternatives for local people and reduced international demand needed; overall loss continues	9
	Model answer - Level 3 (8 marks)	
	Protected areas are the core of rainforest conservation. Korup National Park in Cameroon protects 1260 km ² with about 600 tree species, including the endemic <i>Ancistrocladus korupensis</i> whose alkaloids have anti-HIV and antimalarial activity; Brazil's Tumucumaque National Park covers 39 000 km ² and adjoins a reserve in French Guiana; the Rio Bravo area in Belize is funded by charities, companies, ecotourism and selective logging of high-value mahogany that does not damage the forest long term. Debt-for-nature swaps have cancelled Guatemalan and Brazilian debt in return for protection, CITES listing restricts trade in Honduras mahogany, the ITTO sets sustainable timber aims, ecotourism and community employment give local people an income from standing forest, and carbon sequestration payments are emerging.	
	Where they are enforced, protected areas work, but they face illegal logging, mining, poaching, road building and conflict, and ecotourism itself has damaged sites in Costa Rica and Malaysia. The deeper problem is that the drivers of deforestation - population growth and international demand for beef, soya, palm oil, timber and minerals - originate outside the forest, and climate change threatens even well-protected areas; the ITTO's aims are largely unmet and overall loss continues.	

Question	Marking guidance	Marks
	The most effective schemes are those that combine large protected areas with economic alternatives for local people, such as Rio Bravo, but they can only slow the loss unless demand in importing countries also falls.	
	Weaker answer - Level 2 (4 marks)	
	Rainforests are protected by national parks where logging is banned, for example in Brazil. Debt-for-nature swaps mean a country's debt is cancelled if it protects its forest. Ecotourism brings money from tourists so the forest is worth more standing than cut down. Sustainable logging only takes some trees. These efforts have saved some areas but deforestation is still happening because of illegal logging and farming, so they have only been partly effective.	
	What makes the difference	
	• Level 2 covers the main types of effort and a fair conclusion, but the question says 'describe' - Level 3 gives named examples with figures (Korup, Tumucumaque, Rio Bravo) as the mark scheme expects.	
	• Level 3 explains <i>why</i> effectiveness is limited (drivers originate outside the forest, ITTO aims unmet, climate change) rather than just noting continued loss.	
	• The judgement in Level 3 identifies what works best and what would be needed; Level 2 stops at 'partly effective'.	
10.2	<i>1 mark.</i> • Carbon sequestration reducing climate change; genetic resources / medicines; rainfall regulation; global biodiversity (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 56-57; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
11.1	Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Temperate broadleaf woodland: seasonality, deep soils, high biodiversity, resources, climate and soil regulation; deforestation, fragmentation, management change; SSSIs, ancient woodland protection, coppicing, mixed planting • Tropical rainforest: stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils; fuelwood, timber, agriculture, mining, reservoirs, climate change; protected areas, debt-for-nature swaps, sustainable exploitation • Tropical coral reefs: cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity; fisheries, erosion protection, medicines, tourism; damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species; MPAs, no-take zones, control of activities • Deep-water reefs: cold, dark, slow growth; research, fisheries; trawling, oil / gas, acidification; SACs • Oceanic islands: isolation, endemism, no mammal predators; exploitation, introduced species, habitat change, sea level; eradication, control of development and visitors • Mangroves: halophytes, anaerobic mud; erosion protection, fisheries nurseries, sediment trapping; clearance for aquaculture / urban / ports, pollution, climate change; protection, replanting • Antarctica: extreme cold, seasonality, marine nutrients, ice; water store, albedo, carbon, research; climate change, ozone, tourism, fishing, minerals; Antarctic Treaty, CCAMLR, tourism and waste rules • Synthesis: features -> importance -> threats -> strategy; evaluation of success; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why monitor: species present, population size, density, distribution, survival, age structure, breeding rates, movements; baseline and evaluation; Red List categorisation and quotas rely on it • Standard techniques and their limitations: quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR; sampling design (random / systematic, size, number, timing, standardisation) • New technologies: satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers • How they improve validity: representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity • Examples: penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked • Limitations: cost, battery / tag weight, data handling, ethics, still need ground truthing • Judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7.	25